

1. In a throw of three dice the probability that at least one die shows up 1 is
(a) $5/6$ (b) $91/216$
(c) $1/36$ (d) $125/216$
2. The probability that a marksman will hit a target is given as $1/5$. Then his probability of at least one hit in 10 shots is
(a) $1 - \left(\frac{4}{5}\right)^{10}$ (b) $\frac{1}{5^{10}}$
(c) $1 - \frac{1}{5^{10}}$ (d) None of these
3. Four coins are tossed. The probability that at least one head turns up, is
(a) $1/16$ (b) $1/4$
(c) $15/16$ (d) None
4. The event A is independent of itself if and only if $P(A) =$
(a) 0 (b) 1
(c) 0 or 1 (d) None
5. A dice is thrown $(2n + 1)$ times. The probability of getting 1, 3 or 4 at most n times, is
(a) $1/2$ (b) $1/3$
(c) $1/4$ (d) None
6. The binomial distribution for which mean = 6 and variance = 2 is
(a) $\left(\frac{2}{3} + \frac{1}{3}\right)^6$ (b) $\left(\frac{2}{3} + \frac{1}{3}\right)^9$
(c) $\left(\frac{1}{3} + \frac{2}{3}\right)^6$ (d) $\left(\frac{1}{3} + \frac{2}{3}\right)^9$
7. There are n letters and n addressed envelopes. The probability that each letter takes place in right n envelop is
(a) $\frac{1}{n!}$ (b) $\frac{1}{(n-1)!}$
(c) $1 - \frac{1}{n!}$ (d) None
8. Two squares are chosen at random on a chess-board. The probability that they have a side in common, is
(a) $1/9$ (b) $2/7$
(c) $1/18$ (d) $1/1024$
9. Word **UNIVERSITY** is arranged randomly. Then the probability that both 'I' does not come together, is
(a) $3/5$ (b) $2/5$
(c) $4/5$ (d) $1/5$
10. A dice is thrown two times. If getting the odd number is considered as success, then the probability of two success is
(a) $1/2$ (b) $1/4$
(c) $3/4$ (d) $2/3$
11. In a horse race the odds in favour of three horses are 1:2, 1:3 and 1:4. The probability that one of the horse will win the race is
(a) $37/60$ (b) $47/60$
(c) $1/4$ (d) $3/5$
12. A fair coin is tossed 100 times. The probability of getting tails an odd number of times is
(a) $1/2$ (b) $1/8$
(c) $3/8$ (d) None
13. If A and B are two independent events such that $P(A) = 1/2$, $P(B) = 1/5$ Then
(a) $P\left(\frac{A}{B}\right) = \frac{1}{2}$ (b) $P\left(\frac{A}{A \cup B}\right) = \frac{5}{6}$
(c) $P\left(\frac{A \cap B}{A' \cup B'}\right) = 0$ (d) All of the above
14. A bag contains 6 red, 5 white and 4 black balls. Two balls are drawn. The probability that none of them is red, is
(a) $12/13$ (b) $24/70$
(c) $11/35$ (d) $6/35$
15. In a single throw of one die the probability of obtaining 3 and 4, is
(a) $6/36$ (b) $7/36$
(c) $12/36$ (d) None
16. Out of 21 tickets marked with numbers from 1 to 21, three are drawn at random. The chance that the numbers on them are in A.P, is
(a) $10/133$ (b) $9/133$
(c) $9/1330$ (d) $10/1330$
17. A card is drawn at random from a pack of 100 cards numbered 1 to 100. The probability of drawing a number which is a square is
(a) $1/5$ (b) $2/5$
(c) $1/10$ (d) None
18. For any event A
(a) $P(A) + P(\overline{A}) = 0$ (b) $P(A) + P(\overline{A}) = 1$
(c) $P(A) > 1$ (d) $P(\overline{A}) < 1$
19. A die is rolled three times, the probability of getting a larger number than the previous number each time is
(a) $15/216$ (b) $5/54$
(c) $13/216$ (d) $1/18$

20. If A and B are two independent events, then A and B are
(a) Not independent (b) Also independent
(c) Mutually exclusive (d) None
21. A six faced die is so biased that it is twice as likely to show an even number as an odd number when thrown. It is thrown twice. The probability that the sum of two numbers thrown is even, is
(a) $1/12$ (b) $1/6$
(c) $1/3$ (d) $5/9$
22. A pack of playing cards was found to contain only 51 cards. If the first 13 cards which are examined are all red, then the probability that the missing cards is black, is
(a) $1/3$ (b) $2/3$
(c) $1/2$ (d) $\frac{{}^{25}C_{13}}{{}^{51}C_{13}}$
23. Three at the six vertices of a regular hexagene are chosen at random. The probability that the triangle with these three vertices is equilateral, is equal to
(a) $1/2$ (b) $1/5$
(c) $1/10$ (d) $1/20$
24. If $P(A) = 0.65$, $P(B) = 0.15$, then $P(\overline{A}) + P(\overline{B}) =$
(a) 1.5 (b) 1.2
(c) 0.8 (d) None of these
25. Two dice are tossed. The probability that the total score is a prime number is
(a) $1/6$ (b) $5/12$
(c) $1/2$ (d) None
26. A single letter is selected at random from the word 'SAURABH'. The probability that the selected letter is a vowel is
(a) $4/11$ (b) $4/7$
(c) $3/7$ (d) $3/14$
27. If a dice is thrown twice, then. The probability of getting 1 in the first throw only is
(a) $1/36$ (b) $3/36$
(c) $5/36$ (d) $1/6$
28. The chance of fail in physics are 20% and the chance to fail in mathematics are 10%. What are the chances to fail in at least one subject
(a) 28% (b) 38%
(c) 72% (d) 82%
29. In a throw of a die what is the probability of getting a number less than 7
(a) 0 (b) 1
(c) $1/2$ (d) None
30. Six boys and six girls Sit in a row what is the probability that the boys and girls sit alternatively
(a) $1/462$ (b) $1/2$
(c) $1/924$ (d) None

Rough Works

SCORE SHEET

a b c d	a b c d	a b c d	a b c d	a b c d	a b c d
1. ☐ ☐ ☐ ☐	6. ☐ ☐ ☐ ☐	11. ☐ ☐ ☐ ☐	16. ☐ ☐ ☐ ☐	21. ☐ ☐ ☐ ☐	26. ☐ ☐ ☐ ☐
2. ☐ ☐ ☐ ☐	7. ☐ ☐ ☐ ☐	12. ☐ ☐ ☐ ☐	17. ☐ ☐ ☐ ☐	22. ☐ ☐ ☐ ☐	27. ☐ ☐ ☐ ☐
3. ☐ ☐ ☐ ☐	8. ☐ ☐ ☐ ☐	13. ☐ ☐ ☐ ☐	18. ☐ ☐ ☐ ☐	23. ☐ ☐ ☐ ☐	28. ☐ ☐ ☐ ☐
4. ☐ ☐ ☐ ☐	9. ☐ ☐ ☐ ☐	14. ☐ ☐ ☐ ☐	19. ☐ ☐ ☐ ☐	24. ☐ ☐ ☐ ☐	29. ☐ ☐ ☐ ☐
5. ☐ ☐ ☐ ☐	10. ☐ ☐ ☐ ☐	15. ☐ ☐ ☐ ☐	20. ☐ ☐ ☐ ☐	25. ☐ ☐ ☐ ☐	30. ☐ ☐ ☐ ☐

PROBABILITY - 04

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| 1. (b) | 2. (a) | 3. (c) | 4. (c) | 5. (a) | 6. (c) | 7. (a) | 8. (c) | 9. (c) | 10. (b) |
| 11. (d) | 12. (a) | 13. (d) | 14. (b) | 15. (d) | 16. (c) | 17. (c) | 18. (b) | 19. () | 20. (b) |
| 21. (d) | 22. (c) | 23. (b) | 24. (b) | 25. (b) | 26. (c) | 27. (d) | 28. (a) | 29. (b) | 30. (d) |